

**BRAC University**  
**Department of Electrical and Electronic Engineering**  
**Lab Project**  
**EEE 101L / ECE 101L – Electrical Circuits I Laboratory**  
**Total Marks: 100**

**Project Title: Design of a Two-Source Voltage Monitoring Circuit**

**1. Background**

Some electrical systems can have several operating states even when only one point is available for measurement. A simple way to identify the state is to produce a different DC output voltage for each condition. By measuring this voltage, the user can determine which sources are ON or OFF. In this project, you will design this type of circuit using only resistors and two DC voltage sources.

**2. Objective**

Design, simulate, build, and test a passive resistive circuit that produces a different output voltage for each ON/OFF combination of two DC voltage sources.

**3. Problem Statement**

Each group will be assigned two source voltages and two target output voltages. Design a resistor network with one output terminal. The output must represent the four possible source conditions shown below using four clearly different voltage levels.

| State | Source 1 | Source 2 |
|-------|----------|----------|
| S0    | OFF      | OFF      |
| S1    | ON       | OFF      |
| S2    | OFF      | ON       |
| S3    | ON       | ON       |

The output will be measured using a digital multimeter. First, measure the output without an external load. Then connect the specified 10 k $\Omega$  monitoring load and measure the output again. Students must determine the circuit arrangement and resistor values and support the final design with theoretical calculations, simulation results, and hardware measurements.

**4. Definitions**

- ON: the branch is connected to its assigned source voltage.
- OFF: the branch is connected to ground (0 V). The branch must not be left disconnected.
- No-load output: no external resistor is connected across the output; only the digital multimeter or simulation voltage marker is connected.
- Loaded output: the 10 k $\Omega$  monitoring resistor is connected between the output terminal and common ground.
- Output voltage: the voltage measured between the output terminal and common ground.

## 5. Design Specifications

### Specification 1 - Individual-Source Output Levels

For State S1 and State S2 under no-load conditions, the output voltage must be within  $\pm 5\%$  of the corresponding target assigned to the group.

### Specification 2 - Combined-Source Output

For State S3 under no-load conditions, use the resistor values of the final design to calculate the separate no-load contributions of Source 1 and Source 2. Then add these two contributions using the Superposition Theorem to obtain the expected S3 output. Do not calculate the S3 output by directly adding the assigned S1 and S2 target values. The simulated and measured S3 output must be within  $\pm 5\%$  of the calculated value.

### Specification 3 - Distinguishable Loaded States

With the 10 k $\Omega$  monitoring load connected, list the four output voltages from lowest to highest. The difference between each pair of neighboring voltage levels must be at least 0.50 V so that the four states can be distinguished clearly.

### Specification 4 - Components and Operating Limits

- Use only the two assigned DC voltage sources, resistors, switches, wires, and the 10 k $\Omega$  monitoring resistor. Do not use active devices, capacitors, or inductors.
- Both sources must share the same ground reference. Neither source may exceed 12 V.
- Use standard resistor values between 1 k $\Omega$  and 10 k $\Omega$ . You may use series or parallel resistor combinations when necessary.
- The magnitude of the current in each source branch must remain below 10 mA in every operating state.

### Specification 5 - Implementation Consistency

Use the same nominal resistor values in the simulation and the hardware circuit. If the design is changed, update both versions and explain the change in the report.

## 6. Groupwise Parameters

| Group | Source 1, V1 (V) | Source 2, V2 (V) | S1 target (V) | S2 target (V) |
|-------|------------------|------------------|---------------|---------------|
| 1     | 10               | 6                | 2.2           | 1.1           |
| 2     | 9                | 5                | 2.0           | 1.0           |
| 3     | 12               | 7                | 2.6           | 1.4           |
| 4     | 8                | 4                | 1.6           | 0.8           |
| 5     | 11               | 6                | 2.0           | 1.4           |
| 6     | 10               | 5                | 2.3           | 1.0           |

No target value is given for S3. Each group must calculate it from the two no-load contributions produced by its final resistor network, as described in Specification 2.

## 7. Project Tasks

### Task 1 - Design and Analysis

Choose a suitable resistor network arrangement and practical resistor values. Use appropriate analysis methods learned in this course to calculate the no-load and loaded output voltages, source currents, and resistor power dissipation. Briefly explain your design choices and any changes made when an early design did not meet all requirements.

### Task 2 - Simulation

Implement the proposed circuit in PSpice and evaluate all four source states, both without and with the 10 k $\Omega$  monitoring load. Use the simulation results to check the specifications and revise the design where necessary.

### Task 3 - Hardware Implementation

Build the final circuit on a breadboard using the same nominal component values as the simulation. Measure the actual component values, test all required operating conditions, and record any problems found during testing and how they were corrected.

### Task 4 - Validation

Compare theoretically calculated, simulated, and measured results. Determine whether every specification is satisfied. Explain important differences caused by resistor tolerance, measurement error, wiring problems, or the effect of the digital multimeter on the circuit.

## 8. Deliverables

| Item | Deliverable                    | Required evidence                                                                                                                       |
|------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Group report                   | One PDF report prepared using the template in Section 9.                                                                                |
| 2    | Simulation files               | Complete PSpice files required to reproduce the results.                                                                                |
| 3    | Hardware prototype             | One functioning breadboard circuit for demonstration and assessment.                                                                    |
| 4    | Design evidence                | Circuit schematic, calculations, simulation outputs, hardware photographs, measured data, and a summary table of components with price. |
| 5    | Presentation and demonstration | A short group presentation explaining the design, results, and working hardware circuit.                                                |

## **9. Project Report Template**

Use the following headings in the submitted report. The report should present the group's own design process and evidence; it should not merely repeat the project statement.

### **1. Cover Page**

- Course code and section; project title and group number; student names and IDs; submission date.

### **2. Assigned Requirements**

- Summarize the assigned source voltages, target levels, definitions, and constraints in your own words.

### **3. Design Development**

- Present the final circuit diagram with labels and component values. Briefly mention any other resistor values or circuit arrangements that you tried, why they did not meet all the requirements, and why you selected the final design.

### **4. Theoretical Analysis**

- Show sufficient analysis to justify the predicted outputs. Show calculations for the output voltages both without and with the 10 k $\Omega$  monitoring resistor, the magnitude of current in each source branch, and the power dissipated by the resistors.

### **5. Simulation**

- State whether you used PSpice, explain how you represented the four ON/OFF states, and list the simulation settings used. Include a readable schematic and clearly labelled results for every required state, with and without the monitoring load.

### **6. Hardware Implementation**

- List the nominal and measured component values. Include a clear photograph of the breadboard, describe how the circuit was tested, and summarize any problems found and the changes made to correct them.

### **7. Results and Discussion**

- Compare target or predicted, calculated, simulated, and measured results. State whether every specification was satisfied and explain important discrepancies.

### **8. Conclusion**

- Summarize the final design performance and the main engineering decisions made.

### **9. Appendix**

- Include the group contribution statement and any supplementary calculations or results.

## 10. Assessment Rubrics

### CO4 - Simulation (30 marks)

| Performance Criteria Descriptor                              | Marks     |
|--------------------------------------------------------------|-----------|
| Design for simulation                                        | 10        |
| System/circuit development in simulation environment         | 10        |
| Selection of appropriate simulation parameters               | 5         |
| Validation through simulated data/graphs of input and output | 5         |
| <b>Total</b>                                                 | <b>30</b> |

### Hardware Implementation (30 marks)

| Performance Criteria Descriptor                          | Marks     |
|----------------------------------------------------------|-----------|
| Selection of appropriate components                      | 7         |
| Construction of circuit based on selected solution       | 15        |
| Accuracy in data collection from the constructed circuit | 8         |
| <b>Total</b>                                             | <b>30</b> |

### Project Report (20 marks)

| Assessment Criterion                                                 | Marks     |
|----------------------------------------------------------------------|-----------|
| Completeness and organization                                        | 4         |
| Circuit diagram and theoretical calculations                         | 5         |
| Results and comparison of calculated, simulated, and measured values | 5         |
| Discussion of errors, design choices, and conclusion                 | 4         |
| Clarity, formatting, and group-contribution statement                | 2         |
| <b>Total</b>                                                         | <b>20</b> |

### Presentation (20 marks)

| Assessment Criterion                                 | Marks     |
|------------------------------------------------------|-----------|
| Clear explanation of the problem and final design    | 6         |
| Demonstration of the circuit and key results         | 6         |
| Understanding of the work and responses to questions | 5         |
| Participation of group members and time management   | 3         |
| <b>Total</b>                                         | <b>20</b> |

## 11. CO-PO Mapping

| Course Outcome | PO | Applicable Indicator | Knowledge Profile | Complex Engineering Problem Attributes |
|----------------|----|----------------------|-------------------|----------------------------------------|
| CO4            | e  | e3                   | K6                | P1, P3, P7                             |